SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech AIDS / 3rd Year	**Date:**	20/08/2026

Code of Conduct

I Abinesh H (192424387) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________Abinesh H________

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near…” and “book a flight from Chennai to…”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	4	4	3	3	18	92
2	4	3	4	4	4	19	
3	4	3	3	4	4	18	
4	4	3	4	4	4	19	
5	4	4	3	3	4	18	

Faculty In-charge	Course Coordinator	Program Director
Dr. Kumargurubaran T		
Signature: _______________	Signature: _______________	Signature: _______________
Date: 20/08/2026	Date: ___________________	Date: ___________________

1. Constituency Parsing vs. Dependency Parsing

Constituency parsing represents a sentence as a hierarchy of phrases or constituents such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). It is useful for understanding the nested structure of a sentence.

Dependency parsing directly represents relationships between individual words. Each word is connected to its syntactic head, making relations such as subject, object, modifier, and complement explicit.
For long-distance dependencies, dependency parsing is generally more direct because it can connect related words even when they are separated by many other words. For example, in *"Which book did the student say that the teacher recommended?"*, the dependency structure can directly represent the relationship between *book* and *recommended*.

Justification: Dependency parsing is more suitable for conversational NLP when the main goal is extracting meaningful word-to-word relationships. It is especially useful for information extraction, question answering, intent detection, and semantic analysis. Constituency parsing remains valuable when detailed phrase hierarchy is required.

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2. Top-Down Parsing vs. Earley Parsing

Conventional top-down parsing begins with the start symbol and attempts to generate the input sentence by expanding grammar rules. It works well for complete and relatively simple inputs but can encounter difficulties when the input is incomplete or ambiguous.

Earley parsing is a chart-parsing algorithm capable of handling **context-free grammars**, including ambiguous grammars. It maintains partial parsing states, allowing it to process prefixes of sentences without requiring the entire input to be complete.
For example, when a user types *"best hotels near…"*, Earley parsing can maintain possible partial interpretations and predict what grammatical structures may follow. Similarly, *"book a flight from Chennai to…"* can be parsed incrementally while waiting for the destination.

Justification: Earley parsing is more appropriate for interactive NLP because it supports incomplete input, ambiguity, and incremental parsing. This makes it useful for search suggestions, autocomplete, conversational interfaces, and voice assistants.

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3. CFG vs. PCFG vs. Neural Constituency Parsers

A **Context-Free Grammar (CFG)** represents syntactic structures using deterministic grammar rules. For an ambiguous sentence, CFG can produce multiple valid parse trees but does not inherently determine which interpretation is most likely.

A **Probabilistic Context-Free Grammar (PCFG)** extends CFG by assigning probabilities to grammar rules. When multiple parse trees are possible, the system can select the tree with the highest probability. Thus, PCFG provides a principled way to resolve structural ambiguity while retaining interpretable grammar rules.

Neural constituency parsers use machine-learning models, often based on Transformer architectures, to learn syntactic patterns from large annotated datasets. They can capture complex linguistic patterns and usually provide higher practical accuracy than traditional grammar-based approaches. However, their decisions are less interpretable.

Approach	Ambiguity Handling	Interpretability	Practical Accuracy
CFG	Generates multiple parses	High	Moderate
PCFG	Uses probabilities to select likely parse	High	Good
Neural Parser	Learns likely structures from data	Lower	Very high

Justification: A **PCFG** provides the best balance between interpretability and practical accuracy when explainable grammar checking is important. If maximum accuracy is the primary objective and sufficient training data is available, a neural constituency parser would generally be preferred.

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4. Feature Structures vs. Traditional CFG

Traditional **Context-Free Grammar (CFG)** mainly represents syntactic structures through production rules such as:
S → NP VP
However, CFG by itself does not easily enforce grammatical properties such as **number, gender, person, and tense**.
Feature structures extend grammar rules by associating linguistic features with words or constituents. For example:
NP: [number = singular, gender = feminine]
A verb can then be required to have matching features:
VP: [number = singular]

This allows the system to detect agreement errors such as:
She go to school.
because *she* requires a singular verb form, while *go* is not compatible with the required features.
Feature structures can represent:

- **Number:** singular/plural
- **Gender:** masculine/feminine/neuter
- **Person:** first/second/third
- **Tense:** past/present/future
- **Case:** nominative/accusative, etc.

Justification: Feature structures are particularly useful in multilingual NLP because different languages have different agreement and grammatical constraints. They provide a flexible mechanism for enforcing these constraints while retaining the structural advantages of grammar-based parsing.

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5. Subcategorization Frames

Subcategorization frames specify the types and number of arguments that a verb requires. Different verbs have different argument structures.
For example:

Verb	Subcategorization Frame	Example
give	NP + NP + PP	Give the book to John
sleep	No obligatory object	John sleeps
put	NP + PP	Put the file on the desk

The verb **give** normally requires a giver, an object being given, and a recipient. **Sleep** does not require an object. **Put** requires both an object and a location.
Subcategorization information helps a parser determine whether a sentence is structurally valid. For example:
John put the file.
is incomplete in the usual sense because *put* requires a location or destination complement.
It also helps semantic analysis by identifying **semantic roles**, such as agent, object, recipient, and location.

Justification: Subcategorization is important because it connects **syntactic structure with semantic interpretation**. In voice assistants, it helps determine what entities a command refers to and what actions should be performed, improving intent recognition and accurate command execution.